

DS 412 : Matrix Computations for Data Science

Course Code : DS 412

Course Name : Matrix Computations for Data Science

L-T-P-C : 3-0-2-4

Intended for : Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines

Prerequisite : IC 110-Engineering Mathematics, IC 111-Linear Algebra; From 2022 UG Batches

Mutual Exclusion:

Approval: 55th BoA

Course Contents

- **Basics:** Errors in numerical computations. Review matrices and transformations, Matrix and Vector Norms. (2 lectures)
- **Matrix factorizations:** Cholesky factorization, QR factorization, Householder reflectors and Givens rotations. LU factorization and Gaussian elimination. Numerical stability Pivoting strategies and solution of triangular and full systems by factorization. (5 lectures)
- **Eigenvalue problem:** Basic theory, Power method, bisection method, QR algorithm. Similarity reduction. (4 lectures)
- **Linear least squares problem:** Singular value decomposition, Moore-Penrose Pseudoinverse. Perturbation theory. Least square and normal equations. SVD and rank deficiency, Principal Component Analysis, Linear Discriminant Analysis. (6 lectures)
- **Iterative methods for linear systems :** Iterative methods for linear systems Iterative methods: Jacobi, Gauss-Seidel and SOR iterations. Kronecker product. Krylov subspace methods, conjugate gradient method, preconditioning. (5 lectures)
- **Sparse and banded linear systems:** Storage schemes for banded and sparse matrices, Sparse matrices and sparse solutions: approximate inverses, eigenvalues, incomplete factorization. Matrix regularization: matrix completion. (6 lectures)

Lab Exercises:

Lab would be conducted on a 2-hour weekly slot. Lab sessions would be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class.

Textbooks:

1. Trefethen, Lloyd N., and David Bau III, **Numerical linear algebra**, Vol. 50, SIAM, 1997.
2. Eldén, Lars., **Matrix methods in data mining and pattern recognition**, Vol. 4. SIAM, 2007.



References:

1. Watkins, David S., **Fundamentals of matrix computations**, Vol. 64, John Wiley & Sons, 2004.
2. Demmel, James W., **Applied numerical linear algebra**, Vol. 56, SIAM, 1997.
3. Golub, Gene H., and Charles F. Van Loan., **Matrix computations**, Vol. 3, JHU press, 2012.
4. Cullen, Charles G., **An introduction to numerical linear algebra**, PWS Publishing Company, 1994.